

BigCorp Platform Architecture Overview

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Version: 3.1 | Last Updated: April 2026

System Overview

BigCorp's platform is built on a microservices architecture running on AWS EKS (Kubernetes). We operate 28 production services across 3 AWS regions (ap-south-1, us-east-1, eu-west-1). Traffic is routed via Cloudflare CDN and AWS ALB. Service-to-service communication uses gRPC internally and REST externally.

Core Services

Service	Language	Database	Owner
user-service	Go	PostgreSQL	User Platform
auth-service	Go	Redis + Postgres	User Platform
payment-service	Java	PostgreSQL	Payments
order-service	Python	PostgreSQL	Marketplace
catalog-service	Python	MongoDB + ES	Marketplace
search-service	Python	Elasticsearch	Marketplace
notification-svc	Node.js	Redis	User Platform
ml-serving	Python	Redis (cache)	ML Platform
analytics-ingester	Go	Kafka + S3	Data Platform

Data Infrastructure

Our data stack follows a medallion architecture (bronze-silver-gold):

- Bronze: Raw events land in S3 via Kafka Connect (Confluent Cloud)
- Silver: Spark jobs (EMR Serverless) clean and deduplicate
- Gold: dbt models in Snowflake serve analytics and ML
- Feature Store: Feast on Redis (online) + S3/Parquet (offline)
- Orchestration: Airflow on EKS (managed via Helm)

Observability Stack

Layer	Tool	Retention
Metrics	DataDog	15 months
Logs	DataDog Logs	30 days hot, 1yr cold
Traces	DataDog APM	15 days

Alerts	PagerDuty	N/A
Uptime	DataDog Synthetics	N/A
Error Tracking	Sentry	90 days

Security Architecture

Authentication: OAuth 2.0 via Auth0 (customer-facing) and Okta (internal SSO). Secrets management: AWS Secrets Manager + Sealed Secrets for K8s. Network: VPC with private subnets, NAT gateway, no public IPs on services. WAF: Cloudflare WAF (OWASP Top 10 rules) + AWS WAF on ALB. Encryption: TLS 1.3 in transit, AES-256 at rest (KMS-managed keys). Access control: IAM roles with least-privilege, reviewed quarterly.